Cell mediated cytotoxic reaction

The 1st set of rejection of allograft is brought about mainly by CMI response. In this process the cells involved in the cytotoxic mediated immunity involves. On stimulation of these cells interferon causes the lysis of the graft.

Antibody mediated cytotoxic reaction

The 2nd set rejection of graft is brought about mainly by HMI response. This is one of the hyperacute rejection brought about by the antibodies. Complement, macrophages, mast cells, platelets, B cells bring about this reaction.

Graft versus Host Rejection (GVH)

Sometimes the graft tissue elicits an immune response against the host antigens. This immune response is called graft versus host reaction. It occurs when:

- Graft remains inside the host and the host should not reject the graft.
- The graft should have immune competent T cells.
- The transplantation antigens of the host should be different from that of the graft.



A transplanted heart usually beats slightly faster than normal because the heart nerves are cut during surgery.

Mechanism of the graft

The graft lymphocytes aggregate in the host lymphoid organs and are stimulated by the lymphocytes of the host. The stimulated lymphocytes produce lymphokines. Lymphocytes in turn activate the host T cell. Activated T cell further activates the B cells. The stimulated B cell reacts with the self antigen and causes the damage.

How to prevent graft rejection?

Before transplantation the following things should be done to avoid graft rejection.

- Perform blood grouping and Rh grouping
- HLA typing should be done
- Immuno suppressive drugs should be administered
- Suitable graft should be used

11.5 Immunization/Vaccination

Father of Immunization is Edward Jenner. He produced the vaccine for small pox from cow pox virus.Vaccine is a substance that is introduced into the body to prevent the disease produced by certain pathogens. Vaccines consist of dead pathogens or live but attenuated (artificially weakened) organisms.

Immunization programmes and the development of new vaccines play an important role in protecting individuals against illness. Vaccination works by safely exposing individuals to a specific pathogenic microbe, artificially increasing their immunity to it.

Vaccines are made from

- Live micro-organisms that have been 'treated' so that they are weakened (attenuated) and are unable to cause disease.
- Dead micro-organisms.
- Some part or product of the micro-organism that can produce an immune response.

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